Jintao Li 李锦韬

PhD Researcher · Intelligent Optimization and Decision-Making

Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China

j.t.li@i4ai.org
github.com/Fican1
fican1.github.io
ORCID 0000-0003-0403-6767

I study intelligent optimization for complex decision problems. My work draws on reinforcement learning and evolutionary computation, with an emphasis on multi-objective and constrained optimization. I apply these methods to analog electronic design automation and large-model post-training.

RESEARCH FOCUS

Intelligent Optimization

Learning-based search for expensive problems with competing objectives and hard constraints, using reinforcement learning and evolutionary computation.

Analog EDA

Circuit sizing, behavior-centric representation, PVT-robust design, and industrial EDA integration.

Large-Model Post-Training

Comparison feedback, Pareto-front improvement, preference optimization, and constrained policy learning.

EDUCATION & RESEARCH EXPERIENCE

2024.09–2027.06 **PhD Researcher**, Shenzhen Institute for Advanced Study, UESTC. Expected completion: June 2027.

2021.09–Present **Prof. Yun Li's Research Group**, UESTC. Research on intelligent optimization, especially reinforcement learning and evolutionary methods for multi-objective problems, with applications in analog EDA and large-model post-training.

2021.09–2024.06 **MEng, Electronic Information**, Shenzhen Institute for Advanced Study, UESTC.

2017.09–2021.06 **BEng, Optoelectronic Information Science and Engineering**, Guangzhou University.

SELECTED PUBLICATIONS (1–5)

1. **Learning from Comparison: Constrained Projection Policy Optimization for Pareto-Front Improvement.**
   Jintao Li, Maowen Tang, Yongji Long, Weixuan Liu, Yanlang Zheng, Sicheng He, Ao-Jin Li, Shui Yu, Yun Li. **ICML 2026 · CCF-A.**
2. **Priority-Based Graph-Enhanced Reinforcement Learning for Robust Analog Circuit Optimization.**
   Jintao Li, Zhenxin Chen, Sicheng He, Ao-Jin Li, Shui Yu. **AAAI 2026 · CCF-A.**
3. **Decoupling Analog Circuit Representation from Technology for Behavior-Centric Optimization.**
   Jintao Li, Haochang Zhi, Jiang Xiao, Keren Zhu, Yun Li. **DAC 2025 · CCF-A / EDA flagship.**
4. **Knowledge Transfer Framework for PVT Robustness in Analog Integrated Circuits.**
   Jintao Li, Yanhan Zeng, Haochang Zhi, Jingci Yang, Weiwei Shan, Yongfu Li, Yun Li. **IEEE TCAS-I 71(5), 2017–2030, 2024.**
5. **Multitask Evolution with Problem Reformulation for Global Exploration in Analog Circuit Design.**
   Jintao Li, Sicheng He, Aojin Li, Shui Yu, Yun Li. **Advanced Engineering Informatics 70, 104195, 2026.**

SELECTED PUBLICATIONS (6–10)

6. **Frequency-Domain Driven Recurrent Attention with Linear Complexity for Time Series Forecasting.**
   Jintao Li, Hui Deng, Jiang Xiao, Aojin Li, Sichen He, Shui Yu. **ICASSP 2026, Barcelona.**
7. **AnalogGym: An Open and Practical Testing Suite for Analog Circuit Synthesis.**
   Jintao Li, Haochang Zhi, Ruiyu Lyu, Wangzhen Li, Zhaori Bi, Keren Zhu, Yanhan Zeng, Weiwei Shan, Changhao Yan, Fan Yang, Yun Li, Xuan Zeng. **ICCAD 2024 · EDA flagship.**
8. **Multi-Task Evolutionary to PVT Knowledge Transfer for Analog Integrated Circuit Optimization.**
   Jintao Li, Haochang Zhi, Weiwei Shan, Yongfu Li, Yanhan Zeng, Yun Li. **ICCAD 2023 · EDA flagship.**
9. **Balancing Objective Optimization and Constraint Satisfaction for Robust Analog Circuit Optimization.**
   Jintao Li, Haochang Zhi, Jiang Xiao, Yanhan Zeng, Weiwei Shan, Yun Li. **ASP-DAC 2025, 190–196.**
10. **Robust Circuit Optimization under PVT Variations via Weight Optimization Problem Reformulation.**
    Jintao Li, Yongfu Li, Yanhan Zeng. **Expert Systems with Applications 248, 123301, 2024.**

RESEARCH SYSTEMS

Analog-IC EDA Software Stack — Lead Developer
A mature workflow connecting a server-side optimizer, a 100-GB multimodal circuit database, a lightweight GUI, and industrial simulation tools.

3 circuit classes · 4 tapeout validations · 22 topologies · 4 technology nodes

Self-Evolving Analog Circuit Design Agent — Ongoing Team Project
An agentic framework for specification extraction, behavioral equation derivation and repair, optimizer configuration, simulation execution, human review, and persistent design memory.

Document agent · Modeling agent · Evaluation agent · Human-in-the-loop sign-off

AnalogGym — Open Benchmark
An open testing suite for reproducible evaluation of analog synthesis methods across five circuit categories, multiple process technologies, and both commercial and open-source simulators.

Paper: arXiv 2409.08534 · Code: github.com/CODA-Team/AnalogGym

HONORS & AWARDS

2025	**National Scholarship for Graduate Students**, UESTC.
2025	**CAST Young Elite Scientist Sponsorship Program — Doctoral Track**.
2025	**UESTC Outstanding Student Nominee**, selected among 20 university-wide candidates.
2023–2025	**First-Class Scholarship**, UESTC.
2023	**Outstanding Graduate Student Scholarship**, UESTC.
2022	**Second-Class Scholarship**, UESTC.

RESEARCH LINKS

Google Scholar
scholar.google.com · Jintao Li

ResearchGate
researchgate.net/profile/Jintao-Li-17

ORCID
0000-0003-0403-6767

GitHub
github.com/Fican1